

Atharva Tiwari

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EDUCATION

Worcester Polytechnic Institute Bachelor of Science in Industrial Engineering; Minor in Computer Science	Worcester, MA December 2024
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Worcester Polytechnic Institute Master of Science in Computer Science	Worcester, MA Expected Dec 2026
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WORK EXPERIENCE

ABB Data Science Intern	Cary, NC May 2025 – Feb 2026
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- Developed a multi-class classification model to predict root causes from customer complaints using NLP and structured engineering data, achieving ~90% accuracy and reducing resolution time.
- Designed and implemented a scalable machine learning architecture leveraging GitHub and Kubernetes, for future team adoption.
- Fine-tuned domain-specific embedding models on engineering text to detect inconsistencies between EBOM and quote data.
- Applied NLP and statistical analysis to uncover insights from customer complaints, driving data-informed products and service enhancements.

GE Aerospace Supply Chain Intern	Lynn, MA May 2023 – Aug 2023
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- Developed a software-system to streamline and acknowledge plant-wide CNC programming requests, enhancing efficiency and response times by reducing lead time of request completion by 2 weeks.
- Automated the generation of daily quality dashboards using a VBA macro in Excel, reducing reporting time from 1 hour to 5 minutes and increasing the team's focus on high-value tasks.

Eaton Corporation Operations Management Intern	Milwaukee, WI May 2022 – Aug 2022
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- Automated SAP Material List generation to reduce cycle times and increased material selection accuracy by 30%.
- Developed a real-time Power BI dashboard for production tracking, reducing downtime by 30 minutes.

PROJECT AND RESEARCH EXPERIENCE

Continuous-Time Modeling for Knowledge Tracing using Neural ODEs – Master's Thesis

- Implementing and analyzing Neural ODEs, Augmented Neural ODEs (ANODEs), ODE-RNNs, and Liquid Time-Constant Networks (LTCs) on synthetic and benchmark datasets.
- Studying expressivity, stability, and training dynamics of continuous-depth models compared to discrete-time baselines.
- Exploring the applicability of Neural ODEs for student learning dynamics modeling, with focus on temporal generalization and interpretability.

Unveiling Communication and Support Dynamics in Conflict Zones – Major Qualifying Project

- Developed an ML model using LDA to classify helpline messages with 70% accuracy, enabling targeted resource allocation.
- Conducted Exploratory Data Analysis (EDA) to generate actionable insights for NGO staff and helpline management.
- Conceptualized resource allocation recommendations based on the results from the ML analysis and EDA.
- Designed an ML model to predict vulnerabilities and quantify the impact of key factors on the protection of Ukrainian citizens abroad.

Multi-modal Stock Prediction – Using Sentiment and Historical Data to Predict Stock Price Direction

- Designed a multi-modal deep learning architecture combining FinBERT-based news sentiment with Transformer-based time series modeling for stock prediction.
- Implemented cross-modal feature fusion to integrate textual and temporal signals.
- Benchmarked against baseline models (LSTM, Transformer, sentiment-only), achieving significant performance gains on next-week direction prediction.

AI Powered Whiteboard for Virtual Collaboration

- Designed and implemented an AI-powered whiteboard system using Google Media Pipe Hands for real-time hand gesture recognition, enabling touchless, intuitive interaction in virtual environments like Zoom and Microsoft Teams
- Benchmarked SOTA models to evaluate trade-offs between accuracy and computational efficiency in low-resource settings

SKILLS

Programming: Python, C++, JavaScript, SQL

AI/ML: PyTorch, Scikit-Learn, NLP, LLM Fine-tuning, Transformers, Neural ODEs, Liquid Time-Constant Networks, Agentic AI

Data & Analytics: Pandas, NumPy, Data Analysis, Visualization

DevOps: Git, Azure, AWS, Docker, Experiment Tracking

Mathematics & Foundations: Linux, Statistics, Linear Algebra, Optimization

RELEVANT COURSEWORK

Machine Learning, Deep Learning, Generative Artificial Intelligence, Prescriptive Analytics, Stochastic Processes, Operating Systems